

Find a polynomial $p(x) = \sum_{k=0}^n a_k x^k$ of degree n no higher than 4, such that $p(0) = 1$, $p(1) = 2$, $p(2) = -3$ and $\sum_{k=0}^n a_k^2$ is minimal

The following set of equations must hold true for the polynomial $p(x)$

x	$a_4 x^4$	$a_3 x^3$	$a_2 x^2$	$a_1 x$	a_0	$p(x)$
0	0	0	0	0	1	1
1	1	1	1	1	1	2
2	16	8	4	2	1	-3

Written in matrix form as $A\hat{v} = \vec{b}$

$$\begin{pmatrix} 0 & 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 16 & 8 & 4 & 2 & 1 \end{pmatrix} \cdot \begin{pmatrix} a_4 \\ \vdots \\ a_0 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix}$$

Where we are looking for the vector $\hat{v}$ which has the smallest Euclidian norm. This can be achieved using (as is written in the course material)

$$\hat{v} = A^T(AA^T)^{-1}\vec{b}$$

for vector $\vec{w}$ given by $\vec{w} = (AA^T)^{-1}\vec{b}$

$$\hat{v} = A^T\vec{w}$$

Considering

$$AA^T = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 5 & 31 \\ 1 & 31 & 341 \end{pmatrix}$$

And that $AA^T\vec{w} = \vec{b}$ we get

$$\begin{pmatrix} 1 & 1 & 1 \\ 1 & 5 & 31 \\ 1 & 31 & 341 \end{pmatrix} \vec{w} = \begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix} \implies \vec{w} = \begin{pmatrix} 1/10 \\ 1 \\ -1/10 \end{pmatrix}$$

From which we get

$$\hat{v} = 1/10 \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} + 1 \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} - 1/10 \begin{pmatrix} 16 \\ 8 \\ 4 \\ 2 \\ 1 \end{pmatrix} = \begin{pmatrix} -3/5 \\ 1/5 \\ 3/5 \\ 4/5 \\ 1 \end{pmatrix}$$

And therefore $p(x) = \frac{-3}{5}x^4 + \frac{1}{5}x^3 + \frac{3}{5}x^2 + \frac{4}{5}x + 1$

Q. E. D.